FIRST ORDER DIFFERENTIAL EQUATIONS PRACTICE QUESTIONS

Part A — Separable Differential Equations

1. Solve:

$$\frac{dy}{dx} = xy$$

2. Solve the IVP:

$$\frac{dy}{dx} = \frac{y}{x}, \qquad y(1) = 3.$$

3. Solve:

$$\frac{dy}{dx} = \frac{x^2}{1+y^2}$$

4. Solve:

$$(1+y^2)\,dy = x\,dx$$

5. Solve the growth model:

$$\frac{dP}{dt} = 5P.$$

Part B — Linear First-Order DEs

6. Solve:

$$\frac{dy}{dx} + 3y = 6.$$

7. Solve the IVP:

$$\frac{dy}{dx} - 2y = e^{3x}, \qquad y(0) = 1.$$

8. Solve using integrating factor:

$$x\frac{dy}{dx} + y = x^2.$$

9. Solve:

$$\frac{dy}{dx} + \frac{1}{x}y = 4x.$$

10. Solve:

$$\frac{dy}{dx} + 2xy = x.$$

Part C — Exact Equations

11. Solve:

$$(2xy + 3)\,dx + (x^2 + 2y)\,dy = 0.$$

12. Determine whether the DE is exact. If so, solve it:

$$(3x^2y + 2y)\,dx + (x^3 + 2x)\,dy = 0.$$

13. Check if exact; if not, find an integrating factor depending on x:

$$(2xy + y^2)\,dx + (x^2 + xy)\,dy = 0.$$

Part D — Integrating Factor (Non-Exact Linear Equations)

14. Solve:

$$\frac{dy}{dx} + \left(\frac{2}{x}\right) y = 3x^2.$$

15. Solve:

$$\frac{dy}{dx} - y\cot(x) = \sin(x).$$

16. Solve the IVP:

$$\frac{dy}{dx} + 4y = e^{-2x}, \quad y(0) = 0.$$